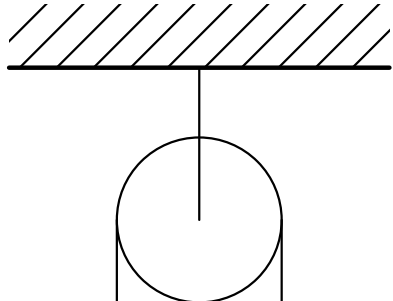
Problem FoL.1 ... (hopefully just) easy pulleys

3 point

Suppose that we have a pulley and some weights hanging from a cord as shown in the figure. Assume that both the cord and the pulley are ideal and massless, $M = 2.0\,\mathrm{kg}$ and $m = 1.0\,\mathrm{kg}$. Find the acceleration of the weight M (in the downwards direction). *Lego wants to discover the simplest pulley problem that half of all teams still fail.*



Problem FoL.2 ... two circuits

3 point

Legolas found several old identical resistors and connected them in series. Such a circuit ha a total resistance $R_\mathrm{S} = 10.0\,\mathrm{k\Omega}$. After that, he decided to connect the resistors in parallel an measured a resistance $R_\mathrm{P} = 1.0\,\Omega$. What was the resistance of a single resistor?

Lego made himself a lego out of resistors

Problem FoL.3 ... felling trees

3 point

A coniferous tree could be (from a mechanical point of view) approximated by a homogeneou right circular cone with height $h = 40\,\mathrm{m}$ and radius at the base $r = 1.0\,\mathrm{m}$. Find the maxima angle by which its axis may be displaced from the vertical axis before it starts to fall due to it weight.

Dodo was procrastinating on Youtube

Problem FoL.4 ... the nearest asteroid

3 point

On 16th August 2020, an asteroid (later called 2020 QG) has been recorded as the closes asteroid (spotted so far) that flew by the Earth without colliding with it. At the nearest poin of approach, it was only $2\,950\,\mathrm{km}$ *above Earth's surface and it had a velocity* $v = 12.3\,\mathrm{km\cdot s^{-1}}$ *How much higher was its velocity compared to the escape velocity at that height above Earth' surface? Find the ratio* v/v_{esc}*.* *Karel made a problem out of news from the website astro.cz*

Problem FoL.5 ... unstable

3 point

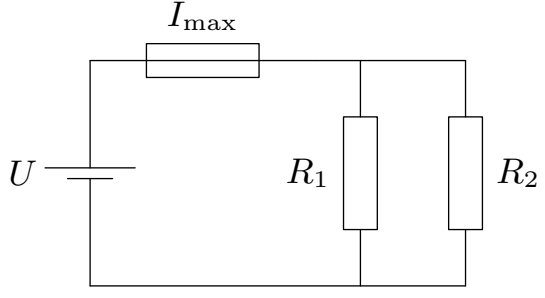
Suppose that we have a cuboid with dimensions $a = 20\,\mathrm{cm}$, $b = 30\,\mathrm{cm}$, $c = 50\,\mathrm{cm}$ *and density* $\rho =$ $= 620\,\mathrm{kg{\cdot}m^{-3}}$. *One of its faces with dimensions* a *and* c *is lying on a horizontal surface i* *homogeneous gravitational field. How stable is it with respect to rotation around one o* *ts bottom edges with length* c*? Find the smallest amount of energy needed to turn it over* *Assume* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Danka's stuff was falling down

Problem FoL.6 ... safe power

3 point

There are two appliances with resistances $R_1 = 500\,\Omega$ and $R_2 = 2\,000\,\Omega$ respectively, connected in parallel to the terminals of a battery through a fuse with a maximum allowed current $I_{\max} = 500\,\mathrm{mA}$. What is the maximum power we could get from the circuit? *Dodo must pay attention at the dormitory.*



Problem FoL.7 ... the ink-blooded prince

3 point

When Harry Potter did not behave, professor Umbridge would punish him by forcing him t write with his own blood. He needed $1.21\,\mu\mathrm{l}$ *of blood on average to write one lowercase letter A book of school rules has* 259 *standard pages, each of which contains* 1 488 *letters on average However, each thirty-seventh letter is a capital letter, which means it consumes three time more ink than a lowercase letter. Assuming that Harry has* $5\,\mathrm{l}$ *of blood, how many copies of th rules could be made from him?* *Jáchym found the whole book series considerably illogica*

Problem FoL.8 ... slowed down train

4 point

While arriving at a station, a train decelerates evenly. Its braking distance is $s = 75\,\mathrm{m}$ *an*
during the penultimate (second to last) second before stopping, it drives a distance $l = 2.25\,\mathrm{m}$
What is its initial velocity v_0 *before it begins to brake?*

Verča took advantage of a train delay to think of new problems

Problem FoL.9 ... we pump oil

4 point

Find the efficiency of a pump, if the electric current flowing through it has an effective value $I =$ $= 125\,\mathrm{mA}$ *when attached to a standard power supply of voltage* $U = 230\,\mathrm{V}$. *Our pump pump* *il with density* $\rho = 870\,\mathrm{kg{\cdot}m^{-3}}$ *through pressure difference* $\Delta p = 120\,\mathrm{hPa}$ *with volumetric flo* *ate* $Q = 0.83\,\mathrm{ml{\cdot}s^{-1}}$.

Dodo wanted to get a rotary vane pump

Problem FoL.10 ... homogeneous air

4 point

From the point of view of classical optics, every substance is homogeneous. However, today w know that everything consists of particles. Determine the number of particles (molecules) of ai that are, under standard conditions, contained in a cube, where the length of one edge of thi cube corresponds to the wavelength of the yellow D-line of sodium.

Dodo couldn't sleep again

Problem FoL.11 ... our old clock

4 point

Our old clock is battery-powered and the capacity of the battery is $E = 1.2\,\mathrm{Wh}$. *The hou hand is* $r_\mathrm{h} = 5\,\mathrm{cm}$ *long and the minute hand is* $r_\mathrm{m} = 8\,\mathrm{cm}$ *long. The linear density of each han is* $\tau = 10\,\mathrm{g{\cdot}m^{-1}}$. *The efficiency of the clockwork is* $\eta = 7\,\%$. *How long does it take for the cloc to stop? The hands rotate continuously and any potential energy released during descending i lost.*

Michal was late due to a delayed clock

Problem FoL.12 ... cough cough

4 point

A deadly virus with a long incubation period has spread around the world and already infecte one ten-thousandth of the total population. Hard-working researchers haven't found any cur yet, but managed to develop a test which can decide whether a given person is infectious or not t returns the positive result in 99.99 % of infected cases. However, it returns the false positiv esult in 0.03 % healthy cases. It may seem that the test is quite reliable. We have chosen an ested a random person. The test returned a positive value. Estimate the probability that th hosen person is infected.

Matěj felt sick during statistical physics practice

Problem FoL.13 ... uneven illumination

4 point

Danka was sitting at a round table with a radius $R = 1.0\,\mathrm{m}$ *and noticed that the edge of th*
able was illuminated much less than the center. What is the difference between illuminance i
he center and at the edge of the table if the only light source in the room is a light bulb wit
uminous intensity $I = 120\,\mathrm{cd}$ *hanging* $h = 1.5\,\mathrm{m}$ *above the center of the table? The light bul*
s an isotropic source of light and the ceiling is black. *Danka couldn't see her books wel*

Problem FoL.14 ... autumn in a train

4 point

A train is climbing an icy track uphill. The steepest section of track which the train is still abl to climb has a slope angle $\alpha = 1.75°$. When the train gets over the slope, it reaches a statio where tracks are horizontal and still icy. What is the shortest distance at which the train ca stop from a speed $v = 52\,\mathrm{km{\cdot}h^{-1}}$?

Dodo was waiting for a train

Problem FoL.15 ... the power of a waterfall

4 point

A $h = 30\,\mathrm{m}$ *high waterfall has flow rate* $Q = 1.2\,\mathrm{m}^3\cdot\mathrm{s}^{-1}$. *Find the total force with which wate impacts the ground under the waterfall. Assume that the water quickly flows away from th point of impact and the depth of water under the waterfall is negligible.*

Dodo is reminiscing about National park Plitvička jezera

Problem FoL.16 ... burning coal

4 point

We burn $m = 213\,\mathrm{mg}$ of pure carbon in a closed vessel of volume $V = 20\,\mathrm{l}$ filled with air. Onc
he temperature equilibrium between the vessel and its surroundings is restored, we measur
he pressure in the vessel. Find the ratio of the pressure after burning to the pressure befor
urning.

Dodo wanted to be malicious

Problem FoL.17 ... heat and phases

4 point

How much higher is the amount of heat required to let ice boil away (Q_b) compared to th amount of heat necessary to only melt it (Q_m)? Find $k = Q_\mathrm{b}/Q_\mathrm{m}$. Our ice is taken fron fridge with inner temperature $t = -18\,^\circ\mathrm{C}$. The specific heat capacity of liquid water is c = $= 4\,180\,\mathrm{J\cdot kg^{-1}\cdot K^{-1}}$ and that of ice is $c_0 = 2\,090\,\mathrm{J\cdot kg^{-1}\cdot K^{-1}}$. The enthalpy of fusion of ic is $l_1 = 334\,\mathrm{kJ\cdot kg^{-1}}$ and the enthalpy of vaporization of water is $l_2 = 2.26\,\mathrm{MJ\cdot kg^{-1}}$. We ar interested in the points when the ice fully melts to water (at $0\,^\circ\mathrm{C}$) and when the water full vaporizes (at $100\,^\circ\mathrm{C}$). Karel was wondering about the possibility of scalding above a kettle

Problem FoL.18 ... boiling

4 point

Danka put $V = 1.5\,\mathrm{l}$ of water with a temperature $T_0 = 25\,^{\circ}\mathrm{C}$ in a pot and set it to boil o a stove with wattage (electric power consumption) $P = 1\,200\,\mathrm{W}$. The efficiency of heating th pot with water (i.e. how much heat is transferred to the pot with water) is $\eta = 0.69$ and the hea capacity of the pot is $C = 500\,\mathrm{J\cdot K^{-1}}$. How long does Danka need to wait till the water starts t boil? Find the necessary constants in tables. Danka had to wait a long time while cooking

Problem FoL.19 ... internal resistance

4 point

Consider two identical resistors connected in series to a non-ideal voltage source. In this case the efficiency of the voltage source is 0.87*. Find the efficiency of the voltage source, i.e. the ratio between the power consumed by our resistors and the total power supplied by the source when the two resistors are connected in parallel instead. We assume model the non-ideal voltage source like an ideal voltage source (not a current source) with an internal resistor connected in series.*

Matěj wanted to save some money – so he tried saving electricity

Problem FoL.20 ... raft voyage

4 point

A raft consists of twelve cylinder-shaped wooden logs. Each log is $l = 8.00\,\mathrm{m}$ *long and ha* *radius* $r = 12.0\,\mathrm{cm}$. *When loaded with* $m = 70\,\mathrm{kg}$ *freight, it floats in such a way that the log* *tick* $s = 3.0\,\mathrm{cm}$ *above the water level. What is the density of the wood?*

Jarda was wondering about possible improvements for a trip on wate

Problem FoL.21 ... mysterious ball

4 point

We have a ball with radius $r = 2.0\,\mathrm{cm}$, *which has an unknown mass and is made from an un known material, hanging from a massless spring. When submerged in water with density* $\rho =$ $= 1\,000\,\mathrm{kg}\cdot\mathrm{m}^{-3}$, *the elongation of the spring drops to* 80% *of its value before the ball i submerged. What is the mass of the ball?*

Verča was reminiscing about experiments in high schoo

Problem FoL.22 ... illuminated

5 point

Danka was sitting in a futuristic train, which was traveling at a constant speed $v = 1.0{\cdot}10^5\,\mathrm{km{\cdot}s}$
The train track was perpendicular to a long straight road and along this road, lamps were place
with constant spacing $l = 5.0\,\mathrm{m}$. *Danka crossed the intersection of the track and the road. I*
he landscape's reference frame, all the lamps were turned on at once when the train was a
distance $d_0 = 2.0\,\mathrm{km}$ *from the intersection. One lamp is at the intersection, let's give i*
ndex 0. *What time passes in Danka's reference frame between the moments when light fro*
amp 0 *reaches her and when light from a lamp with index* 100 *reaches her? Use the exact valu*
f speed of light.

Danka was traveling by train

Problem FoL.23 ... a ball

5 point

Anička is playing with a ball of yarn with a radius $R = 5\,\mathrm{cm}$, which is formed by a $l = 100\,\mathrm{m}$ long string. She stands on an inclined plane with a slope angle $\alpha = 3°$. She kicks the ball along the slope upwards, but one end of the string stays stuck at the point where she kicked it away so the string is unraveling as the ball travels up the slope until the whole ball unravels and only the straight string remains on the slope. Find the smallest initial velocity of the ball needed to reach the state described above.

Matěj read a children's book

Problem FoL.24 ... a traveler's problem

5 point

A traveler needs to catch his train and does not have much time. The train station is $s = 1\,000\,\mathrm{m}$ *ar from him. Unfortunately, he finds himself in the fields, where he has to force his way ou with velocity* $v_\mathrm{f} = 3.0\,\mathrm{km{\cdot}h^{-1}}$*. There is a road leading to the station as well, and on this roac he could jog with an average velocity* $v_\mathrm{r} = 7.0\,\mathrm{km{\cdot}h^{-1}}$*. However, the road is* $l = 600\,\mathrm{m}$ *far fron him. Find the optimal angle* α *(measured with respect to a perpendicular to the road) suc hat the traveler reaches the station as soon as possible if he starts walking towards the roa t this angle.*

Verča went hiking

Problem FoL.25 ... a different parallel-plate capacitor

5 point

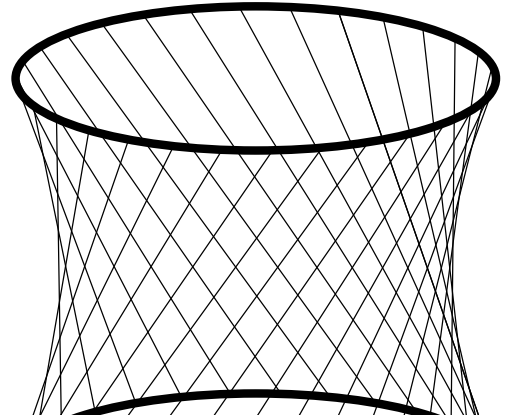
Consider a capacitor consisting of a square conductive plate with surface area $S = 6.0\,\mathrm{cm}$ *and a parallel, infinite and grounded conductive plane at a distance* $d = 1.1\,\mathrm{mm}$. *Find th capacitance of such a capacitor, defined as the ratio of charge on the plate to potential on th plate.* *Originally, Vašek prepared a problem with a capacitor with infinite capacitance*

Problem FoL.26 ... oscillating hoop

5 point

Suppose that we have two identical hoops, each with a radius $R = 1.0\,\mathrm{m}$. *The upper hoop is in a fixed position, while the lower one is attached to the upper one by several massless cords; each of these cords has the same length* $l = 2.0\,\mathrm{m}$ *and hangs vertically. Let the mass of the lower hoop be* $m = 1.0\,\mathrm{kg}$. *If we rotate it a bit around the vertical axis and release, what is its period of small oscillations?*

Lego modified a mathematical problem into a physics problem.



Problem FoL.27 ... oscillating

5 point

Suppose that we have a homogeneous square plate with edge length 1 m. *We drill a tiny hole i*
t, hammer a nail into a wall, hang the plate from the nail through this hole and let the plat
oscillate in the vertical plane in which it lies. How far from the centre of the square do we hav
o drill the hole to maximise the frequency of small oscillations?

Matěj had some spare metal sheet and didn't know what to do with i

Let us use the equation for physical (or *compound*) pendulum

Problem FoL.28 ... lego and dice 1.0

4 point

We have two small cuboids, one lying on the other – the lower cuboid has mass $M = 2.0\,\mathrm{kg}$ *the upper one* $m = 1.0\,\mathrm{kg}$*. The coefficient of friction between the lower cuboid and the surfac under it is* 0*, the coefficient of static friction between the cuboids is* $f_\mathrm{s} = 0.50$ *and the coefficien of kinetic friction between them is* $f_\mathrm{k} = 0.20$*.*

Find the magnitude of the force we must exert on the lower cuboid that causes it to mov with a constant acceleration $a = 10\,\mathrm{m{\cdot}s^{-2}}$*.*

Lego was helping his friend with physics, so he at least took some ideas for new problems

Problem FoL.29 ... lego and dice 2.0

5 point

We have two small cuboids, one lying on the other – the lower cuboid has mass $M = 2.0\,\mathrm{kg}$ *he upper one* $m = 1.0\,\mathrm{kg}$*. The coefficient of friction between the lower cuboid and the surfac nder it is* 0*, the coefficient of static friction between the cuboids is* $f_\mathrm{s} = 0.50$ *and the coefficien f kinetic friction between them is* $f_\mathrm{k} = 0.20$*.*

Find the magnitude of the force we must exert on the lower cuboid that causes it to mov with a constant acceleration $a = 1.0\,\mathrm{m{\cdot}s^{-2}}$*.*

Lego was helping his friend with physics, so he at least took some ideas for new problems

Problem FoL.30 ... impact accelerator

5 point

We have a large number of point masses in a series in one line. The first point mass ha a mass $M_0 = 1\,\mathrm{kg}$ and every subsequent point mass has a mass equal to 70 % of the previou one. The first point mass starts to move towards the second one with a kinetic energy $E_0 = 50\,\mathrm{J}$ All collisions are perfectly elastic. Which point mass will be the first to have a speed large han one percent of the speed of light? Neglect any relativistic effects.

Jarda wanted to improve upon CERN technology

Problem FoL.31 ... jump on!

5 point

magine a sufficiently large cuboid with mass $M = 32.5\,\mathrm{kg}$ *lying on the ground. We throw smaller cuboid with mass* $m = 11.7\,\mathrm{kg}$ *on it in such a way that right before the impact he vertical component of its velocity is almost zero, while the horizontal component is* $v = 19.2\,\mathrm{m{\cdot}s^{-1}}$. *The coefficient of friction between the larger cuboid and the ground is* $f_1 = 0.13$ *he coefficient of friction between the two cuboids is* $f_2 = 0.69$. *What is the total distanc overed by the larger cuboid? Jachym wondered wherever is possible to land with a plane*

Problem FoL.32 ... disgusting piston

5 point

In a thermally insulated cylinder with an internal cross section $S = 500\,\mathrm{cm}^2$ *and a height* $l =$ $= 50\,\mathrm{cm}$, *there is a resistor with resistance* $R = 120\,\Omega$. *The cylinder is otherwise filled with ai at a temperature* $T_0 = 20\,^\circ\mathrm{C}$ *and pressure* $p_0 = 101\,\mathrm{kPa}$, *and the same kind of air surround the cylinder. A current* $I = 200\,\mathrm{mA}$ *flows through the resistor. A base of the cylinder break away when pushed with a force exceeding* $F = 500\,\mathrm{N}$. *After what time does that happen?*

Jarda wanted to break something using air

Problem FoL.33 ... 4D scouring pad

5 point

Find the resistance between two neighbouring vertices of a four-dimensional cube made of wire
Each edge of the cube has a resistance $R = 1\,000\,\Omega$.

Karel was wondering about multi-dimensional budgeting problems

Problem FoL.34 ... levitation by firearm

5 point

Consider a body with mass $M = 12.1\,\mathrm{kg}$, which is kept in the air by sustained fire from AK-47. Calculate the rate of fire necessary for the body to hover and constantly oscillate between two points low above the ground. All the impacts are elastic, the mass of each bullet is $m = 7.93\,\mathrm{g}$ and the bullets are flying straight up with speed $v = 715\,\mathrm{m{\cdot}s^{-1}}$.

Don't forget to take cover and move only when the enemy is reloading

Problem FoL.35 ... burning phosphorus

5 point

Assume that we burn $m = 213\,\mathrm{mg}$ of phosphorus dust in a closed vessel with volume $V = 20$ filled with air. Once the temperature equilibrium between the vessel and its surroundings i restored, we measure the pressure in the vessel. Find the ratio of the pressure after burning t the pressure before burning the phosphorus. *Dodo wanted to be malicious, v. 2*

Problem FoL.36 ... mixed current

4 point

Consider an electronic component which behaves like a resistor with resistance $R = 42\,\Omega$. W
et electric current flow through it; this current has an alternating part and a direct part. Th
lternating part is harmonic with a frequency $f = 50.0\,\mathrm{Hz}$. The minimum value of the curren
s $12\,\mathrm{mA}$ and the maximum value is $42\,\mathrm{mA}$. The current does not change direction. What is th
verage power consumed by the component? *Karel wanted to combine*

Problem FoL.37 ... stopping a train

5 point

There is a train with mass $m = 500\,\mathrm{t}$ *and speed* $v = 100\,\mathrm{km{\cdot}h^{-1}}$ *heading towards a superher*
standing on the tracks. The train is at distance $s = 1\,300\,\mathrm{m}$ *from the superhero, who wants th*
rain to stop just in front of him. However, the only thing available to him is a laser pointe
What power must his laser have to achieve this feat? The front of the locomotive is a mirror.

Jarda thought up the plot for a new film

Problem FoL.38 ... solar cube

5 point

magine that we have a satellite in the shape of a perfect cube with its surface covered by sola
panels. We position it in such a way that it orbits a star with a circular trajectory much large
han the star. The satellite is able to rotate in any way we want. Find the ratio of the maximun
o the minimum radiant power absorbed by the satellite. Assume that all incident radiation i
bsorbed.

Karel saw a solar cube on the Internet

Problem FoL.39 ... optimal direction

6 point

We have a rigid body with mass $M = 1.0\,\mathrm{kg}$ lying on a horizontal surface. The coefficient o
riction between the body and the surface below it is $f = 0.4$. What is the maximum acceleratio
which the body can reach if we exert a force $F = 5\,\mathrm{N}$ on it? Lego was moving boxes

Problem FoL.40 ... watch out, Danka is throwing

6 point

Danka wants to throw a cricket ball on flat ground. Find the launch angle (measured wit respect to the horizontal plane) which she should choose in order to throw the ball as far a possible. Danka is $h = 1.6\,\mathrm{m}$ *tall and throws with a velocity* $v = 4.5\,\mathrm{m{\cdot}s^{-1}}$.

Throwing a cricket ball has never been Danka's favourite spor

Problem FoL.41 ... laser at a dormitory

4 point

Jáchym shines a laser from a dormitory window towards the ground. Matěj measures the radia
ion of the laser on the ground, $\Delta h = 42.0\,\mathrm{m}$ below Jáchym. The laser shines with a frequency f
out Matěj measures a different frequency f. What is the ratio $|f - f_0|\,/f_0$?

Karel repeatedly heard about a black hole being observed

Problem FoL.42 ... pyramid

6 point

Imagine a wooden (density $\rho_w = 600\,\mathrm{kg{\cdot}m^{-3}}$) symmetrical square-base pyramid floating on th surface of a calm lake. A part of the pyramid with height $h = 6.0\,\mathrm{cm}$ is under the water surface We impact the pyramid in the vertical direction. Find the frequency of small oscillations of th pyramid. The density of the water is $\rho_l = 1\,000\,\mathrm{kg{\cdot}m^{-3}}$. Assume that these oscillations do no affect the height of water level in the lake. The pyramid has an ideal shape, it is homogeneou and pointing downwards (i.e. the apex is lower than the base).

Vítek wondered about global warming

Problem FoL.43 ... pole dance

6 point

We have two coaxial cylinders – the smaller one has an outer radius $r = 1.0\,\mathrm{cm}$ *and the bigge one has an inner radius* $R = 10\,\mathrm{cm}$*. The cylinders are connected by a rod positioned in th adial direction. On the rod, next to the surface of the smaller cylinder, there is a small bead There is one more small ball on the surface of the smaller cylinder. We spin the whole systen s a rigid body around the common axis with an angular velocity* $\omega = 2.0\,\mathrm{rad{\cdot}s^{-1}}$ *and the elease the bead and the ball simultaneously. The bead can only move on the rod; the ball ca nove freely. What is the absolute difference between the times when the ball and the bead hi he inner surface of the larger cylinder?*

Jarda got a bit flummoxed from this

Problem FoL.44 ... a fall into the unknown

7 point

We have an infinite grounded conductive plane in free space without any gravitational fields At a distance $d = 1.00\,\mathrm{m}$ from the plane, there is a small ball with mass $m = 2.00\,\mathrm{g}$ and electri charge with magnitude $q = 4.00\,\mu\mathrm{C}$. We release the ball. How long does it take for the charg to drop on the plane?

Jarda knew that it would fall eventually, but he wanted to know when exactly

Problem FoL.45 ... spring of knowledge

7 point

Jáchym owns a well of knowledge, which has depth $h = 32\,\mathrm{m}$ and a constant circular cross ection with radius $r = 1.5\,\mathrm{m}$. At the bottom of the well, in the middle, there is a poin which radiates knowledge uniformly into the whole space around it with power P_0. Knowledg propagates in space similarly to light rays. The reflectance of the sides of the well is $k = 0.42$ The bottom of the well of knowledge does not reflect anything, it absorbs knowledge perfectly Let P denote the total knowledge power radiated out of the well. Find the ratio P/P_0.

Wells are inexhaustible sources of ideas

Problem FoL.46 ... an incredible space battle

8 point

Imagine us in the middle of a fight in space between civilization 1 and civilization 2. A battl cruiser of civilization 1 has just launched a rocket on a battle cruiser of civilization 2. Th cruisers are $s = 5.00\,\mathrm{km}$ *from each other and do not move with respect to each other. Th properties of the rocket are: initial mass* $m_0 = 5.00\,\mathrm{t}$, *engine thrust* $T = 1.50 \cdot 10^5\,\mathrm{N}$, *specifi impulse (exhaust velocity with respect to the rocket)* $u = 3.00\,\mathrm{km{\cdot}s^{-1}}$. *The engines are set t maximum thrust from launch till the moment of impact. Find the velocity of the rocket righ before it crashes into the cruiser of civilization 2.* *Jindra was watching Star Wars*

Problem FoL.47 ... rotational pump

8 point

Vítek would like to get some water from his well, but he does not want to keep pulling th bucket up. Therefore, he gradually stirred the water around faster and faster until, at an angula velocity $\omega = 11\,\mathrm{rad\cdot s^{-1}}$*, the water started flowing out of the well all by itself. Vítek knows th depth of the well (from the top edge to the ground at the bottom)* $h = 47\,\mathrm{m}$*. The well ha a circular cross-section with a radius* $r_0 = 1.6\,\mathrm{m}$*. What was the height of the water colum (from the bottom of the well to the water surface) before Vítek started spinning the wate around?*

Jáchym heard that problems about wells were popular in Fyziklání 2020

Problem FoL.48 ... ideal centrifuge

9 point

magine a tube filled with ideal gas with molar mass $M_m = 36\,\mathrm{g{\cdot}mol^{-1}}$*. The length of the tub*
s $r_0 = 1.00\,\mathrm{m}$*, its thickness is negligible compared to its length. We spin the tube around th*
xis perpendicular to the tube and passing through one of its bases, with an angular velocity ω =
$= 451.00\,\mathrm{s^{-1}}$*. The gas inside will settle in equilibrium, at a temperature* $T = 300\,\mathrm{K}$ *along wit*
he tube. Find the distance between the axis of rotation and the centre of mass of the air i
he tube. The tube is rotating in a horizontal plane. *Jáchym wanted to separate air*

Problem FoL.49 ... falling impulse

9 point

There is a thread with a length $l = 1.00\,\mathrm{m}$ *and a total mass* $m_\mathrm{v} = 1.00\,\mathrm{g}$ *hanging from th ceiling. On its bottom end, a small weight with unknown mass* m *is hung. We send an impuls down the thread from the top and simultaneously drop a point mass from the ceiling. We notic two things:*

1. *the impulse is traveling along the thread with a constant speed,*
2. *the impulse reaches the bottom end of the thread at the same time as the point mass.*

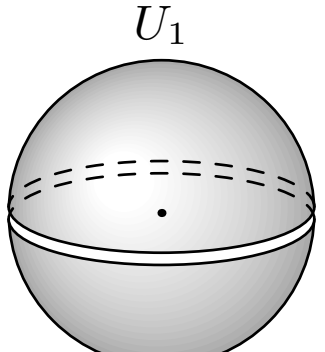
What is the mass of the little weight? Everything is taking place in homogeneous gravity wit acceleration $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$. *Jirka was unable to figure out how this idea struck his mind*

Problem FoL.50 ... a problem with potential

3 point

Lego took two hemispherical shells with the same radii and connected them in a non-conducting way such that together they formed a sphere. Then he set a potential $U_1 = 100\,\mathrm{V}$ on one hemisphere and a potential $U_2 = -100\,\mathrm{V}$ on the other hemisphere. What is the potential in the middle of the sphere?

Lego knew that he has a potential...

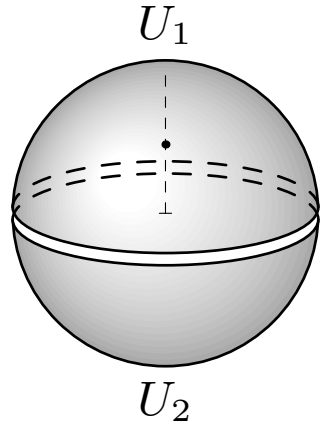


Problem FoL.51 ... another problem with potential

9 point

Lego took two hemispherical shells with the same radii and connected them in a non-conducting way such that together they formed a sphere. Then he set a potential $U_1 = 100\,\mathrm{V}$ on one hemisphere and a potential $U_2 = -100\,\mathrm{V}$ on the other hemisphere. Find the potential in the middle of a straight line connecting the centre of the sphere and the apex of the positively charged hemisphere (depicted in the figure)?

... and he also thought that it is greater than 0.



Problem M.1 ... at the train station

3 point

Legolas was trying to catch a train. He was running with a speed $v = 3.0\,\mathrm{m{\cdot}s^{-1}}$ *when he noticed that one of his shoelaces was untied. Therefore, he stopped and spent* $t = 10.0\,\mathrm{s}$ *tying it. He then jumped on an escalator, which was moving with a speed* $u = 1.0\,\mathrm{m{\cdot}s^{-1}}$ *(on which he ran with speed* v *as well). Suddenly, he facepalmed, realizing he could have made one more step, tied the shoelace on the escalator and saved some time. But how much exactly?*

You do not need to know the exact length of the escalator – it is sufficient to know that it was long enough for Legolas to manage to tie his shoelace on it. *Lego missed the train*

Problem M.2 ... at the airport

3 point

Dano and Danka are in a hurry to catch a plane to Mexico, so they both run on a movin walkway (in the direction of its movement). Both of them have the same step length, whic is exactly 1 m. *Dano runs twice as fast as Danka and he has to take 28 steps to run from on end of the walkway to the other end, while Danka needs to take just 21 steps. How long is th walkway?*

Legolas wanted to catch a plane

Problem M.3 ... under the ground

3 point

Matěj likes to travel by metro. One day, he got off at a station which was $h = 50\,\mathrm{m}$ *deep unde the surface. He walked upwards with a speed* $v = 1\,\mathrm{m{\cdot}s^{-1}}$ *on an escalator, which itself move upwards with a speed* $u = 1.5\,\mathrm{m{\cdot}s^{-1}}$*. What work has Matěj done by walking upwards? Assum that his mass is* $m = 60\,\mathrm{kg}$*.*

Matěj likes escalators and the metro

Problem M.4 ... in the shopping mall

3 point

You have traveled by train, by metro and by plane to visit a shopping mall, which is, of course equipped with many escalators. On a switched-on escalator moving upwards, the time it take you to walk up and down again is $t_{\mathrm{on}} = 50\,\mathrm{s}$. *On a switched-off escalator, it takes* $t_{\mathrm{off}} = 30\,\mathrm{s}$ *What is the speed of the stairs of a switched-on escalator, if you walk with a speed* $v_{\mathrm{u}} = 1.0\,\mathrm{m{\cdot}s^{-1}}$ *upwards and* $v_{\mathrm{d}} = 2.0\,\mathrm{m{\cdot}s^{-1}}$ *downwards?*

Legolas runs up and down escalators in metro stations

Problem E.1 ... statistical 1 - intro

3 point

n statistical physics, we can describe a system of particles enclosed in an unchanging volume We call this a cannonical system and the probability that we find the system in a certain stat follows Boltzmann's Law

$$p_n = \frac{1}{Z} \mathrm{e}^{-\frac{E_n}{k_\mathrm{B} T}} ,$$

where E_n is the energy of the system in the state n, T is the temperature of the system and k_B = $= 1.381 \cdot 10^{-23}\,\mathrm{m^2 \cdot kg \cdot s^{-2} \cdot K^{-1}}$ is the Boltzmann constant. The value of Z is chosen in such a wa hat the sum of p_n over all the states equals one. Typically, there is a large number of state system can reside in, but in this simple problem, let's work with a system with only thre listinguishable states ($n = 1, 2, 3$) with energies $E_1 = 1.00 \cdot 10^{-20}\,\mathrm{J}$, $E_2 = 2E_1$ and $E_3 = 3E_1$ Determine the value of Z at a temperature $T_0 = 275\,\mathrm{K}$.

Matěj missed statistical physics among FYKOS problems

Problem E.2 ... statistical 2 - basics

3 point

Now, let's examine a canonical system which has only two possible states. The difference o
energies between these two states is exactly $\Delta E = 10^{-20}\,\mathrm{J}$. *For example, you can imagine*
molecule that exists either in its ground state or an excited state, but this description works fo
any general two-state system. What is the probability of finding the system in the state wit
higher energy if the temperature of the system is $T_0 = 275\,\mathrm{K}$?

Note: use knowledge from the previous problem. *Because the intro was not enough*

Problem E.3 ... statistical 3 - the quantum way

4 point

Earlier, you had to express Z, which you can calculate using

$$Z = \sum_n \mathrm{e}^{-\frac{E_n}{k_\mathrm{B}T}} ,$$

where the sum goes over all possible states (such as the two in the previous problem). Then you had to use Boltzmann's Law to calculate probabilities. The sum Z is called the partition function.

Now let's consider a more complicated system - the quantum harmonic oscillator, which can be found in many different states. The energy of the n-th state is $\left(n + \frac{1}{2}\right)\hbar\omega$, where $n \in \mathbb{N}$ and $\hbar\omega = 10^{-21}\,\mathrm{J}$ is the parameter of the oscillator. Determine the probability that the system is in the ground state ($n = 0$). The temperature of the system is $T_0 = 275\,\mathrm{K}$.

Hint $\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$ *for* $q \in (0, 1)$. *This is where the fun begins*

Problem E.4 ... statistical 4 - the second quantum one

4 point

In the previous problem, you calculated the partition function of a quantum harmonic oscillato with energies $\left(n + \frac{1}{2}\right)\hbar\omega$, *where* $\hbar\omega = 10^{-21}$ J. *It's actually possible to obtain all the differen thermodynamic properties of the system just using the partition function. For example, th internal energy of the system* $\overline{E}$ *(the expected value of energy) can be calculated as*

$$\overline{E} = k_{\mathrm{B}} T^2 \frac{\partial}{\partial T} \ln\left(Z(T)\right).$$

Determine the internal energy of the quantum harmonic oscillator from the previous problem in the units of $\hbar\omega$. The temperature of the system is $T_0 = 275\,\mathrm{K}$. *This was supposed to be a problem with a second partial derivative, but it would probably be too hard*

Problem X.1 ... the strongman and the weight

3 point

Imagine a $m = 10\,\mathrm{kg}$ weight attached to the end of a $d = 1.0\,\mathrm{m}$ long (and very light) rod. How many times bigger is the force our biceps needs to exert to hold the rod with the weight steady compared to simply holding it in hand? Assume that our forearm is $l = 30\,\mathrm{cm}$ long. We hold the rod in such a way that it forms a "straight-line extension" of our forearm.

Dodo was carrying a pan full of water

Problem X.2 ... (not) breathing

3 point

In a single day, approximately $V = 11\,000\,\mathrm{l}$ of air pass through human lungs. Find the averag
speed of air flowing through the larynx, assuming that its diameter is $d = 40\,\mathrm{mm}$.

Dod and his treacherous factors of two

Problem X.3 ... the pressure of a ballerina

3 point

A ballerina that weighs $m = 50\,\mathrm{kg}$ is landing on a big toe after jumping into the height h = = $0.5\,\mathrm{m}$. At its narrowest point, the last phalanx bone of the big toe has a diameter d = = $1.3\,\mathrm{cm}$. Find the highest stress across this bone during the landing. Assume that the ballerin decelerates upon impact with a constant force for $t = 0.5\,\mathrm{s}$. *Dodo stood on one toe*

Problem X.4 ... saliva

3 point

Fykos organizers are very skillful and managed to create 100 g *a substance such that* 99 % *of it mass is water. Unfortunately, after a while, some water evaporated away and now only* 98 % *o the sample is water. What is the current mass of the sample?* *Lego likes Vsauce2*